

Impact of Enhanced Pharmacologic Care on the Prevention of Falls: A Randomized Controlled Trial

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ABSTRACT

Background: Falls are the leading cause of both fatal and nonfatal injuries among older adults in the United States. Medications that affect the central nervous system are known to increase the risk of falling.

Objective: The purpose of this study was to assess the effects of a community pharmacy-based falls-prevention program targeting high-risk older adults on the rates of recurrent falls, injurious falls, and filling prescriptions for medications that have been associated with an increased risk of falling.

Methods: This was a randomized controlled trial of participants recruited through a community pharmacy chain in North Carolina. The 2-year study consisted of a 1-year “look-back” period before randomization and a 1-year follow-up period after randomization. Patients were eligible to participate if they were ≥65 years of age, had fallen at least once during the 1-year period preceding enrollment, and were taking medications associated with an increased risk of falling. Medications classified as high risk included benzodiazepines, antidepressants, anticonvulsants, sedative hypnotics, opioid analgesics, antipsychotics, and skeletal muscle relaxants. Participants were assigned to either the intervention arm or the control arm; participants in the intervention arm were invited to attend a face-to-face medication consultation conducted by a community pharmacy resident, whereas those in the control arm received no medication consultation. The primary end point was the rate of recurrent falls during the 1-year follow-up period. Secondary end points were the total number of prescriptions for high-risk medications filled during the follow-up period and either discontinued use or a reduction in the dosage of a high-risk medication during the follow-up period.

Results: One hundred eighty-six patients (132 women, 54 men; 88.7% white) were enrolled. Intention-to-treat (ITT) analyses revealed no significant differences in the rates of recurrent falls, injurious falls, or filling prescriptions for high-risk medications. However, 13 patients in the intervention group either discontinued use of a high-risk medication or had the dosage reduced during the follow-up period, compared with 5 patients in the control group ($\chi^2 = 3.94$; $P < 0.05$). As-treated analyses revealed numeric reductions in the rates of falls (rate ratio [RR] = 0.76; 95% CI, 0.53–1.09), injurious falls (RR = 0.67; 95% CI, 0.43–1.05), and filling prescriptions for high-risk medications (RR = 0.85; 95% CI, 0.72–1.03) after receipt of the intervention, but the differences were not statistically significant.

Conclusions: Results of this study support the feasibility of using community pharmacies to deliver a falls-prevention program targeting high-risk older adults. Although the ITT analyses revealed no significant reduction in the rate of recurrent falls, injurious falls, or overall use of high-risk medications, individuals in the intervention group

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were more likely than those in the control group to discontinue use of a high-risk medication or have the dosage reduced during the 1-year follow-up period. More work is needed to evaluate the intervention using a larger sample size that provides greater power to detect clinically meaningful effects of reduction in the use of high-risk medications on preventing or reducing falls in the high-risk population. (*Am J Geriatr Pharmacother*. 2010;8:428–440) © 2010 Elsevier HS Journals, Inc.

Key words: falls, older adults, medications, randomized controlled trial, pharmacy.

INTRODUCTION

Falls are the leading cause of both fatal and nonfatal injuries among adults ≥ 65 years of age in the United States.¹ Although most falls are not fatal, they can contribute to a cascade of events that results in loss of independence and decreased quality of life.^{1–3} Factors that increase the risk of falling include a history of falls; arthritis; depression; gait, balance, or visual deficits; muscle weakness; functional or cognitive impairment; and use of certain medications (eg, antidepressants, benzodiazepines, sedative hypnotics, opioids).^{4–8}

The importance of rationalizing use of medications to reduce the risk of falling is highlighted in the Guideline for the Prevention of Falls in Older Persons, developed jointly by the American Geriatrics Society and the British Geriatrics Society.⁵ Although the effect of specific medications on the risk of falling remains controversial, observational studies have consistently found an increased risk of falling among individuals using medications that affect the central nervous system (CNS).^{8–11} For example, in a 2007 meta-analysis of 98 studies that examined the association between use of CNS-active medications and fractures, Takkouche et al¹¹ reported that the pooled relative risk of fracture associated with use of these medications ranged from 1.15 (95% CI, 0.94–1.39) for hypnotics to 2.17 (95% CI, 1.35–3.50) for barbiturates. In the 10 reviewed studies that assessed the risk associated with use of nonspecified CNS-active drugs, the pooled relative risk for fracture was 1.48 (95% CI, 1.41–1.59). Concurrent use of ≥ 4 prescription medications has also been reported to be an independent predictor of falls, irrespective of the medications used.⁷

A recent Cochrane Review included 111 randomized controlled trials assessing the effectiveness of community-based falls-prevention programs.⁶ However, only 2 studies^{12,13} evaluated the effectiveness of interventions designed to improve medication management. In one study,¹² involving 93 adults ≥ 65 years of

age, gradual withdrawal of psychotropic medications significantly reduced the rate of falls (rate ratio [RR] = 0.34; 95% CI, 0.16–0.73), but not the risk of falling. In the second study,¹³ involving 849 adults ≥ 65 years of age, a medication review and modification program implemented through general practitioners reduced the risk of falling (RR = 0.61; 95% CI, 0.41–0.91). Thus, limited evidence is available to support the effectiveness of interventions designed to improve medication use to reduce the risk of falling among older adults, and more research is needed to verify these early findings.

Over the past 30 years, pharmacists' roles have expanded beyond the distribution of medications to include the provision of comprehensive medication therapy management services.¹⁴ Until recently, much of this role expansion had been limited to pharmacists working in hospitals and outpatient clinics, where they are able to participate on multidisciplinary teams and have access to patients' medical records.^{15,16} Although the role of pharmacists in community settings has evolved more slowly, model community pharmacy-based programs have reported encouraging results in terms of improving patient health outcomes across a variety of disease states, including diabetes mellitus, hypertension, and hypercholesterolemia.^{17–21} However, the authors are not aware of any studies that have evaluated the effects of a community pharmacy-based falls-prevention program.

The purpose of this study was to assess the effects of a community pharmacy-based falls-prevention program targeting high-risk older adults on the rates of recurrent falls, recurrent injurious falls, and filling prescriptions for medications that have been associated with an increased risk of falling.

PATIENTS AND METHODS

Study Design

A complete description of the study has been published previously.²² Briefly, the study was a randomized controlled trial. The 2-year study consisted of a 1-year look-back period before randomization and a 1-year follow-up period after randomization. Participants were recruited from Kerr Drug (Raleigh, North Carolina), a regional chain of ~ 100 community pharmacies located in North Carolina. The study was limited to individuals at high risk for falling, specifically those who were ≥ 65 years of age; had experienced ≥ 1 fall not attributable to syncope within the 1-year period preceding randomization; and were taking ≥ 4 different chronic prescription medications, ≥ 1 of which was a CNS-active medication. Individuals were excluded

from the study if they resided in a long-term care facility; were housebound; were not able to read and write English; or exhibited significant cognitive impairment (≥ 3 errors on a 6-item screening derived from the Mini-Mental State Examination).²³

Participant Recruitment

Potential study participants were identified within a central electronic database of prescription records maintained by Kerr Drug. A letter from Kerr Drug was sent to the identified individuals describing the study and inviting their participation. Individuals who wished to participate in the study or learn more about it were instructed to return a form that was included with the letter. A self-addressed, stamped envelope was included with the letter to facilitate return of the form. Individuals who returned the form were called by study personnel to assess their interest in participating in the study and to determine their eligibility.

Study Participants

Before baseline data were collected, participants were required to return a consent form and HIPAA (Health Insurance Portability and Accountability Act) waiver. The study was approved by the biomedical institutional review board of the University of North Carolina at Chapel Hill.

Intervention Procedures

Immediately after the baseline telephone interview, participants were randomized to either the intervention or control group. Participants assigned to the intervention group received an invitation by telephone to participate in a free, face-to-face medication consultation conducted by a community pharmacy resident at the Kerr Health Care Center nearest their home; participants assigned to the control group received no medication consultation. During the consultation sessions, the pharmacist reviewed the patient's medications and identified potential problems in their drug therapy. Special attention was given to medications that have been found to increase the risk of falling, with an emphasis on CNS-active medications.^{7,8,10,23–26} To standardize delivery of the intervention, structured algorithms for addressing medications associated with a high risk of falling were created by 2 of the study investigators (S.F. and M.T.R.).²² When a drug therapy problem was identified, the pharmacist discussed the problem and potential solutions with the patient. If patients expressed interest in making a change in their medication regimen, the pharmacist contacted their

prescribing physician to inform them of the potential drug therapy problem(s) and seek prescriber approval of the recommended changes. The mean duration of the consultation sessions was 45 minutes.

Participants in both groups received a packet containing 2 brochures on the prevention of falls developed by the Centers for Disease Control and Prevention ("What You Can Do to Prevent Falls" and "Check for Safety: A Home Fall Prevention Checklist for Older Adults"),²⁷ as well as a refrigerator magnet designed for this project, containing contact information for study personnel. Participants were not required to read these materials, and reading of the materials was not assessed.

Data Sources

Telephone Interviews

Baseline data (sociodemographic characteristics, history of falls, health status, and medication use) were collected during the telephone interview. Medication use was assessed using the Brief Medication Questionnaire (BMQ). This instrument contains a 5-item regimen screen that assesses how the medication is used (eg, how many days/week and how many times/day the medication is taken) and a 2-item belief screen that assesses the individual's perception of medication efficacy and bothersome adverse effects.²⁸ The BMQ was readministered at 3-month intervals, ending 12 months after the baseline assessment. In addition, participants were interviewed each time they returned a falls calendar (described below), indicating that they had sustained ≥ 1 fall during the month. These interviews were conducted immediately after each calendar was returned; the discussion centered on the circumstances surrounding the fall(s) and the consequences of the fall(s).

Monthly Falls Calendars

After participating in the baseline telephone interview, each participant received a set of twelve 1-month calendars. They were asked to record each fall they experienced on the calendar for the current month. A *fall* was defined as "a sudden, accidental change in position where you land on the ground, floor, or an object (like a piece of furniture)." Participants were asked to return each calendar to the study staff at the end of that month. Participants who did not return a calendar by the 10th of the following month were called by study personnel to obtain the required information by phone.

Prescription Profile Records

Information concerning prescriptions for high-risk medications filled by study participants was obtained

from the electronic prescription profile records of the Kerr Drug pharmacy from which the participant was recruited. The extracted information included the name of the medication, the quantity and dosage regimen prescribed, the number of days' supply obtained, and the date on which each prescription was filled or refilled. Information was extracted for a 2-year period, beginning 1 year before randomization and ending 1 year after randomization.

End Points

Primary End Point

The primary end point was the rate of recurrent falls during the follow-up period. Data for creation of this end point were derived from the monthly falls calendars. Because falls calendars were mailed to study participants after randomization to their study group, the follow-up period began on the first day of the month after randomization.

Secondary End Points

The study had 2 secondary end points: the total number of prescriptions for high-risk medications filled during the 1-year follow-up period and either discontinued use or reduction in the dosage of a high-risk medication during the 1-year follow-up period. Medications classified as high risk included benzodiazepines, antidepressants, anticonvulsants, sedative hypnotics, opioid analgesics, antipsychotics, and skeletal muscle relaxants.²² The total number of prescriptions for high-risk medications filled during the follow-up period was obtained from participants' prescription profile records at Kerr Drug.

To determine whether any high-risk medications used at baseline were discontinued or reduced in dosage during the 1-year follow-up period, coders (who were masked to the participants' group assignment) reviewed the participants' prescription profile records. A medication was classified as discontinued if all of the following criteria were met: (1) a prescription for the medication was filled at least once ≤ 90 days before randomization; (2) use of the medication at baseline was confirmed by the patient during the baseline interview; (3) no prescriptions for the medication were filled > 120 days after randomization; (4) the medication was not converted from brand name to generic or vice versa; (5) the patient continued to fill prescriptions for other medications at Kerr Drug for the entire follow-up period; and (6) continued use of the medication was not reported in any of the follow-up telephone interviews. The 120-day lag after randomization, during which patients may have filled a prescription for the

high-risk medication, allowed time for the participant to receive the intervention and for any recommended changes in medication regimen to be approved by the participant's physician. Similar criteria were used to identify medications that were reduced in dosage during the follow-up period.

Exploratory End Point

The rate of recurrent injurious falls during the follow-up period was examined as an exploratory end point because it was anticipated that the rate of injurious falls would be too low to provide sufficient power for the primary analyses. Data for creation of this end point were derived from the monthly falls calendars and brief telephone interviews after receipt of a calendar on which a fall was reported. During the interview, participants were asked if they were injured by the fall (yes, no) and, if so, to describe the injuries. Falls were coded as injurious if participants indicated that they were injured by the fall, irrespective of the severity of the injuries.

Power Analyses

Initial power analyses focused on the risk of falling. In these analyses, it was estimated that 262 subjects per group would be required to achieve 80% power to detect a between-group difference of 25% in the risk of falling with an α set at 0.05. Approximately 18 months after initiation of participant recruitment, interim power analyses were conducted when it became apparent that it would be difficult to reach the targeted sample size. At the time of the interim analyses, 75 individuals had been randomized. The interim analyses focused on the rate of recurrent falls rather than the risk of falling. Estimates of the rate of falls were based on the intensity of the recurring event using an Andersen-Gill recurrent event model.²⁹ In these analyses, the hazard ratio (HR) for experiencing a fall was 0.64 (95% CI, 0.33–1.25; $P = \text{NS}$), numerically favoring the intervention group. Based on these results, it was estimated that 95 subjects per group would provide 80% power to detect an HR of 0.64 in the final analyses.

Statistical Analyses

Descriptive statistics were calculated, and associations between participant characteristics and group assignment were assessed using Pearson χ^2 tests, t tests, and Mann-Whitney U tests, depending on the measurement properties of the variable under consideration. Regression analyses were performed in 2 scenarios: intent to treat (ITT) and as treated. In the ITT analyses, outcomes were examined with participants categorized

according to their random group assignment. Thus, individuals were included in the intervention group even if they did not receive the intervention. In the as-treated analyses, participants assigned to the intervention group were classified as control subjects until they received the intervention. Crude incidences of recurrent falls, injurious falls, and filling prescriptions for high-risk medications were estimated under both scenarios, but were not evaluated for statistical significance.

In analyses where the outcome of interest was the rate of recurrent falls, time to the first fall was calculated as the number of days between the date of first follow-up and the date of the fall. For subsequent falls, time to the fall was calculated as the number of days between that fall and the previous fall. Participants who stopped returning calendars before the end of the follow-up period were treated as censored as of the last day on the last calendar they returned. For participants who did not return any calendars, follow-up time was recorded as 0.5 day to prevent them from being excluded from the analyses. In cases in which participants failed to return a calendar for a particular month but returned subsequent ones, the gap in follow-up caused by the missing calendar(s) was removed from computations of follow-up time. Multiple falls that occurred on a single day were recorded as separate falls.

Semiparametric recurrent rate models were used to assess effects of the intervention on the rate of recurrent falls after controlling for potential confounders.³⁰ Four control variables were included in the models because they were associated with the risk of falling in preliminary bivariate analyses: (1) ≥ 2 falls during the year preceding study entry; (2) use of a cane or walker; (3) number of high-risk conditions (dizziness, diabetes mellitus, urinary incontinence, arthritis, Parkinson's disease, or history of stroke); and (4) number of days per week engaged in exercise, sports, or physically active hobbies that increase heart rate or breathing. In addition, age and sex were included in the models because research has consistently suggested that the risk of falling increases with age and that women are more likely than men to be injured in a fall.³¹ These same procedures were used to model the rate of injurious falls.

In analyses where the outcome of interest was the rate of filling prescriptions for high-risk medications, participants who stopped filling prescriptions at the Kerr Drug pharmacy from which they were recruited before the end of the 1-year follow-up period were treated as censored as of the date on which their last prescription was filled, whether or not it was for a high-

risk medication. For participants who did not fill any prescriptions at Kerr Drug during the 1-year follow-up period, follow-up time was recorded as 0.5 day to prevent them from being excluded from the analyses.

In the ITT scenario, a negative binomial regression model was used to assess effects of the intervention on the rate of filling prescriptions for high-risk medications during the follow-up period after controlling for age and sex. The negative binomial model allowed control for overdispersion that was present in these analyses. In the as-treated analyses, each individual who received the medication consultation provided 2 records (before and after the intervention), forming a cluster. A Poisson regression model was fitted with generalized estimating equations (GEEs) using an exchangeable working correlation structure. Overdispersion was managed via the robust variance estimators from GEEs. All regression models controlled for the number of prescriptions for high-risk medications filled during the year before study entry. To minimize bias introduced by an extreme outlier, a log transformation was performed on the variable before including it in the regression models.

Between-group differences in the proportion of participants who discontinued use or reduced the dosage of a high-risk medication were evaluated using χ^2 tests. All analyses were performed using R version 2.81 (R Foundation for Statistical Computing, Vienna, Austria) and SAS version 9.1 (SAS Institute Inc, Cary, North Carolina) software.

RESULTS

As shown in the **figure**, a total of 7793 people were identified from the Kerr Drug pharmacy records and received letters informing them of the study. Of these, 981 returned the form included with the letter indicating interest in the study, 801 agreed to participate in the eligibility interview, 342 were determined to be eligible (2 subsequently declined to participate), and 186 completed baseline data collection procedures and subsequently were randomized to a study group. Characteristics of the 186 study participants are shown in **Table I**. No significant differences were found between the participants in the intervention group and those in the control group regarding the demographic variables assessed, with the exception of sex (78.5% female in the control group vs 63.4% in the intervention group; $P = 0.03$).

Nineteen participants did not return any calendars, 15 participants failed to return a calendar for a particular month but returned subsequent ones, and 9 participants did not fill any prescriptions at Kerr Drug during the

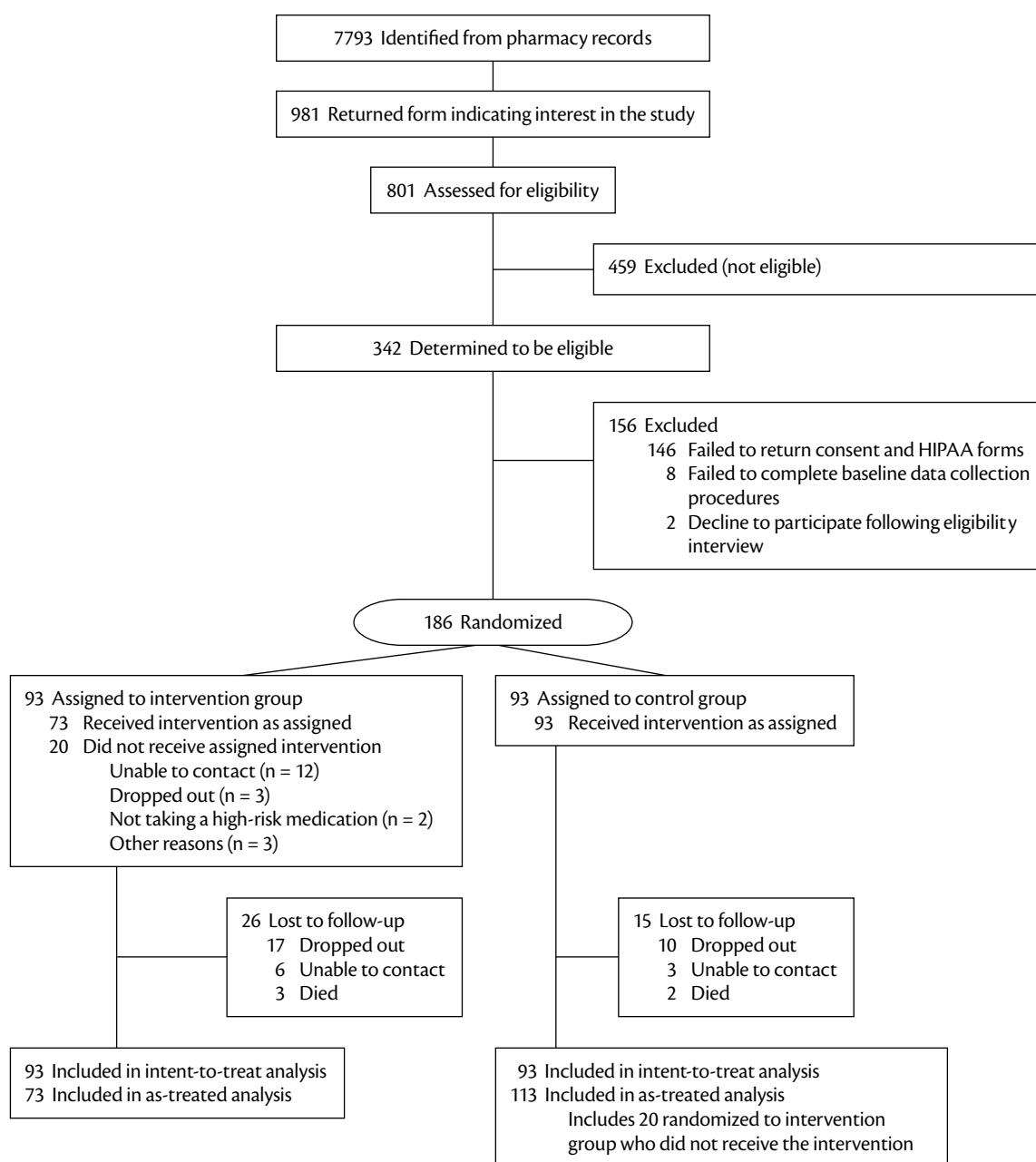


Figure. Flow of participants through the trial. HIPAA = Health Insurance Portability and Accountability Act.

1-year follow-up period. Five patients died during the follow-up period for reasons unrelated to treatment.

Falls

ITT Analyses

In the ITT analyses, individuals were included in the intervention group even if they did not receive the

intervention. Study participants experienced a total of 322 falls during the follow-up period: 53.1% (171/322) occurred among those assigned to the control group and 46.9% (151/322) occurred among those assigned to the intervention group, corresponding to crude incidences of 5.84 and 5.91 per 1000 days of follow-up, respectively; 56.5% (105/186) of the participants ex-

Table I. Demographic characteristics of 186 study participants. Data are presented as number (%), unless indicated otherwise.

Characteristic	Group		P
	Control (n = 93)	Intervention (n = 93)	
Age, mean (SD), y	74.1 (6.8)	75.5 (7.0)	0.17
Sex			0.03
Female	73 (78.5)	59 (63.4)	
Male	20 (21.5)	34 (36.6)	
Race			0.44
White	85 (91.4)	80 (86.0)	
Other	8 (8.6)	13 (14.0)	
Education			0.92
High school or less	22 (23.7)	23 (24.7)	
Some college	24 (25.8)	22 (23.7)	
College degree	45 (48.4)	48 (51.6)	
Unknown	2 (2.2)	0	
Marital status			0.78*
Married	55 (59.1)	55 (59.1)	
Widowed	25 (26.9)	22 (23.7)	
Divorced or separated	13 (14.0)	14 (15.1)	
Never married	0	2 (2.2)	
Retired	77 (82.8)	74 (79.6)	0.58
Insurance other than Medicare	89 (95.7)	82 (88.2)	0.06
No. of high-risk conditions, mean (SD) [†]	1.65 (1.19)	1.58 (1.06)	0.70
No. of exercise days/wk, mean (SD)	3.10 (2.28)	3.19 (2.42)	0.78
Use of cane or walker	37 (39.8)	43 (46.2)	0.37
≥2 Falls during previous year	48 (51.6)	43 (46.2)	0.46
No. of prescriptions for high-risk medications filled during previous year, mean (SD)	14.1 (14.3)	14.3 (11.7)	0.94
Follow-up time for falls, mean (SD), days [‡]	330.4 (106.7)	290.9 (142.1)	0.25
Follow-up time for prescription fills/refills, mean (SD), days [‡]	322.1 (104.1)	318.0 (101.2)	0.78

*Divorced, separated, and never married categories were combined in the statistical test.

[†]Conditions included in the count of high-risk conditions were dizziness, diabetes mellitus, urinary incontinence, arthritis, Parkinson's disease, and history of stroke.

[‡]Times were used in the intent-to-treat analyses.

perienced ≥1 fall during the 1-year follow-up period, 52 in the control group and 53 in the intervention group. After adjusting for covariates (**Table II**), the RR for recurrent falls in the intervention group compared with the control group was 0.96 (95% CI, 0.65–1.40; *P* = NS).

As-Treated Analyses

In the as-treated analyses, participants assigned to the intervention group were classified as control subjects until they received the intervention. Twenty (21.5%) of the 93 individuals randomized to the intervention group did not attend a medication consulta-

tion. Thus, they were treated as control subjects for the entire follow-up period. Three other participants (3.2%) did not return any falls calendars after attending the medication consultation and, thus, were censored in the falls analyses preceding the intervention.

Of the 151 falls experienced by participants assigned to the intervention group, 14 (9.3%) were experienced by individuals who never attended a medication consultation, 23 (15.2%) occurred before the date of the medication consultation, and 114 (75.5%) occurred on or after the date of the medication consultation. In the as-treated analyses, the incidence of falls was 5.97 per 1000 days of follow-up in the control group and 5.71 in the intervention group. After adjusting for covariates (Table II), the RR for recurrent falls in the as-treated analyses was 0.76 (95% CI, 0.53–1.09; $P = \text{NS}$).

Use of High-Risk Medications

Study participants filled a total of 2630 prescriptions for high-risk medications during the year preceding study entry and 1940 prescriptions for high-risk medications during the year after study entry.

ITT Analyses

Of the 1940 prescriptions for high-risk medications filled during the follow-up period, 966 (49.8%) were filled by patients in the control group and 974 (50.2%) by patients in the intervention group, corresponding to crude incidences of 32.6 and 33.5 per 1000 days of follow-up,

respectively. After adjusting for covariates (Table III), the RR for filling a prescription for a high-risk medication in the intervention group compared with the control group was 1.02 (95% CI, 0.82–1.27; $P = \text{NS}$).

In the intervention group, 13 participants either discontinued use of a high-risk medication or had the dosage reduced during the 1-year period after randomization, compared with 5 in the control group ($\chi^2 = 3.94$; $P < 0.05$).

As-Treated Analyses

The incidence of filling a prescription for a high-risk medication was 36.3 and 27.0 per 1000 days of follow-up in the control and intervention groups, respectively. After adjusting for covariates (Table III), the RR for filling a prescription for a high-risk medication in the intervention group, compared with the control group, was 0.85 (95% CI, 0.72–1.03; $P = \text{NS}$).

Among the participants who received the intervention, 16.4% (12/73) either discontinued use of a high-risk medication or had the dosage reduced during the follow-up period, compared with 5.3% (6/113) of those who did not receive the intervention ($\chi^2 = 6.28$; $P < 0.02$).

Injurious Falls

Study participants experienced 127 injurious falls (72 in the control group and 55 in the intervention group). In the ITT analyses, the crude incidence for injurious falls was 2.46 and 2.15 per 1000 days of

Table II. Adjusted rate ratios (with 95% CIs) for recurrent falls from intent-to-treat and as-treated scenarios.

Predictor Variable	Intent to Treat			As Treated		
	Rate Ratio	95% CI	P^*	Rate Ratio	95% CI	P^*
Intervention group [†]	0.96	0.65–1.40	0.75	0.76	0.53–1.09	0.14
Age, y	0.99	0.97–1.02	0.45	0.99	0.97–1.02	0.50
Sex, male	1.50	0.97–2.33	0.07	1.61	1.07–2.43	0.02
≥2 Falls during previous year	2.38	1.72–3.31	<0.01	2.34	1.67–3.26	<0.01
No. of high-risk conditions	1.32	1.16–1.51	<0.01	1.33	1.16–1.52	<0.01
No. of exercise days/wk	1.13	1.04–1.23	<0.01	1.13	1.05–1.22	<0.01
Use of cane or walker	1.95	1.42–2.68	<0.01	1.97	1.44–2.68	<0.01

*Statistical significance of the relationship between the predictor variable and the rate of recurrent falls.

[†]In the intent-to-treat analyses, the number of falls experienced by individuals randomly assigned to the intervention group ($n = 93$) and control group ($n = 93$) were 151 and 171, respectively. In the as-treated analyses, 114 falls were experienced by individuals who participated in the medication consultation intervention ($n = 73$) and followed receipt of the intervention.

Table III. Adjusted rate ratios (with 95% CIs) for filling prescriptions for high-risk medications from intent-to-treat and as-treated scenarios.

Predictor Variable	Intention to Treat*			As Treated†		
	Rate Ratio	95% CI‡	P§	Rate Ratio	95% CI‡	P§
Intervention group	1.02	0.82–1.27	0.87	0.85	0.72–1.03	0.09
Age, y	0.99	0.97–1.00	0.17	0.99	0.98–1.00	0.14
Sex, male	1.10	0.86–1.40	0.46	1.03	0.86–1.24	0.76
No. of prescriptions for high-risk medications filled during previous year (logged)	2.56	2.24–2.93	<0.01	2.59	2.30–2.92	<0.01

*Parameter estimates obtained using negative binomial regression model.

†Parameter estimates obtained using Poisson regression under generalized estimating equations.

‡CIs based on robust variance.

§Statistical significance of the relationship between the predictor variable and the rate of filling prescriptions for high-risk medications.

^{||}In the intent-to-treat analyses, the number of prescriptions for high-risk medications filled by individuals randomly assigned to the intervention group (n = 93) and control group (n = 93) were 974 and 966, respectively. In the as-treated analyses, 553 prescriptions for high-risk medications were filled by individuals who participated in the medication consultation intervention (n = 73) and were filled after receipt of the intervention.

follow-up in the control and intervention groups, respectively; in the as-treated analyses, the crude incidence was 2.46 and 2.06 per 1000 days, respectively. After adjusting for covariates in the ITT analyses, there was an 11% reduction (RR = 0.89; 95% CI, 0.57–1.38; $P = \text{NS}$) in the rate of recurrent injurious falls in the intervention group compared with the control group (Table IV); in the as-treated analyses, there was a 33% reduction (RR = 0.67; 95% CI, 0.43–1.05; $P = \text{NS}$).

DISCUSSION

This study evaluated the effects of a community pharmacy-based falls-prevention program targeting high-risk older adults using a randomized controlled trial design. Analyses based on the ITT scenario revealed no significant differences between the groups in the rate of falls or injurious falls or the rate of filling prescriptions for high-risk medications. However, more than twice as many participants in the intervention group (14.0%; 13/93) discontinued use of a high-risk medication or had the dosage reduced as in the control group (5.4% [5/93]; $\chi^2 = 3.94$; $P < 0.05$). Furthermore, although not statistically significant, results of the as-treated analyses revealed a numeric reduction in the rate of recurrent falls (RR = 0.76; 95% CI, 0.53–1.09), the rate of recurrent injurious falls (RR = 0.67; 95% CI, 0.43–1.05), and the rate of filling prescriptions for high-risk medications (RR = 0.85; 95% CI, 0.72–1.03) after receipt of the intervention.

When interpreting the findings from the present study, one must consider the complexities involved in conducting a trial of this type and the limitations imposed by these complexities. First, considerable difficulty was experienced recruiting a sufficient number of patients to participate in the study, which resulted in the study being underpowered. This experience appeared to be comparable to that observed in other studies engaging older adults in falls-prevention programs, which has proved to be challenging.^{32–36}

Second, even among the individuals who agreed to participate in the study, the intervention was delivered to only 78.5% (73/93) of the participants assigned to the intervention group. For 12 of the participants who did not receive the intervention, either it was not possible to contact the participant to schedule a time for the intervention (despite multiple attempts) or the participant did not attend a scheduled consultation.

Third, only 16.4% (12/73) of the individuals who took part in the medication consultation discontinued use of a high-risk medication or had the dosage reduced during the follow-up period. In some cases, patients may be reluctant to change medications that have already provided relief of symptoms or management of a serious condition, even if therapeutic alternatives are available. In other cases, however, few alternatives may be available or the available alternatives may have already been tried unsuccessfully. Thus, patients' use of medications associated with an increased risk of falling

Table IV. Adjusted rate ratios (with 95% CIs) for recurrent injurious falls from intention-to-treat and as-treated scenarios.

Predictor Variable	Intention to Treat			As Treated		
	Rate Ratio	95% CI	P*	Rate Ratio	95% CI	P*
Intervention group [†]	0.89	0.57–1.38	0.59	0.67	0.43–1.05	0.08
Age, y	1.01	0.98–1.04	0.70	1.01	0.98–1.04	0.66
Sex, male	1.13	0.64–2.00	0.66	1.22	0.72–2.09	0.46
≥2 Falls during previous year	1.89	1.27–2.82	<0.01	1.86	1.24–2.80	<0.01
No. of high-risk conditions	1.30	1.10–1.54	<0.01	1.30	1.10–1.54	<0.01
No. of exercise days/wk	1.11	1.01–1.22	0.04	1.11	1.01–1.22	0.04
Use of cane or walker	1.39	0.89–2.18	0.14	1.41	0.91–2.20	0.12

*Statistical significance of the relationship between the predictor variable and the rate of recurrent injurious falls.

[†]In the intention-to-treat analyses, the number of injurious falls experienced by individuals randomly assigned to the intervention group (n = 93) and control group (n = 93) were 55 and 72, respectively. In the as-treated analyses, 41 injurious falls were experienced by individuals who participated in the medication consultation intervention (n = 73) and followed receipt of the intervention.

may reflect a careful balancing of risks versus benefits, weighted by patient preferences.

Fourth, the delay between randomization and delivery of the intervention adds to the complexity of the study findings. In clinical trials that involve drug testing, exposure to the intervention usually begins immediately after randomization. In the present study, however, time was required after randomization to schedule the medication consultation. Furthermore, in cases where the intervention pharmacist recommended a change in the participant's medication regimen, no change could be made until it was approved by the prescribing physician. This feature of the study design contributed to the difference between findings obtained in the ITT and as-treated scenarios. Almost 25% of the 151 falls experienced by individuals in the intervention group either occurred before the participant received the intervention (23 falls) or were experienced by individuals who never received the intervention (14 falls). Thus, including these falls in the ITT analyses introduced a conservative bias into the study findings. On the other hand, the as-treated analyses resulted in a loss of some of the protection provided by randomization (eg, protection against confounding by indication). An alternative approach, when evaluating a service where (1) there is an anticipated delay between the date of randomization and the delivery of service and (2) the primary outcome involves time-to-event data, would be to identify, a priori, a date to initiate the

follow-up period that allows time for the service to be implemented. In the authors' opinion, if done a priori, this approach would preserve the benefits derived from randomization while avoiding the accumulation of "outcomes" before intervention delivery.

Fifth, medication use was assessed through review of pharmacy prescription records, assuming that study participants took the medications they obtained. This did not allow the authors to assess more fine-grained aspects of medication use such as whether excessive amounts of specific medications were taken on an occasional basis or whether medications were taken at the optimal time of day.

Finally, the generalizability of study findings is limited by the same types of factors that are encountered in other clinical trials (eg, restrictive inclusion and exclusion criteria). More work is needed to evaluate the intervention in a more diverse target population.

Despite these limitations, this study represents an important first step in developing and evaluating a community pharmacy-based falls-prevention program targeting high-risk older adults. Strengths of the study include its randomized controlled design, development and implementation of standardized algorithms to guide intervention delivery, use of community pharmacy residents in an existing residency program to deliver the intervention, examination of medication therapy management as an independent intervention for reducing falls, and rigorous assessment of falls and medication use during a 1-year follow-up period.

In contrast to the 2 randomized controlled trials^{12,13} that reported decreases in the risk or rate of falls after implementation of medication modification interventions, the present study did not require participants to alter their medications¹² and did not utilize participants' physicians to deliver the intervention.¹³ Outside of a research study, it is not feasible to require that patients discontinue the use of certain types of medications, as was done in the study by Campbell et al.¹² However, the effectiveness of the intervention evaluated in the present study might be enhanced by increasing the involvement of patients' physicians. This might be accomplished by delivering the intervention in the context of a collaborative pharmacist–physician practice agreement.

In addition, the extent to which falls and fall-related injuries can be reduced by medication review and modification interventions remains unclear. For example, a recent study³⁷ used a quasi-experimental design to compare changes in the incidence of hip fracture following implementation of a statewide policy in New York that required triplicate forms when benzodiazepines were prescribed. New Jersey served as a comparison state. The study was limited to Medicaid enrollees ≥ 65 years of age. Medicaid and Medicare claims data were analyzed. This study found that use of benzodiazepines decreased by $\sim 60\%$ after implementation of the statewide policy ($P < 0.05$). However, this decline in benzodiazepine use was not accompanied by a decrease in the incidence of hip fracture. Hazard ratios for hip fracture, adjusted for age and Medicaid eligibility category and stratified by patient gender and history of benzodiazepine use, ranged from 0.8 to 1.3 during the 12-month period before implementation of the policy and from 1.1 to 1.3 during the 21-month period after implementation (all, $P = \text{NS}$). Thus, more research is needed to better assess the extent to which the incidence of falls and fall-related injuries can be reduced by interventions targeting high-risk medications.

CONCLUSIONS

Results of this study support the feasibility of using community pharmacies to deliver a falls-prevention program targeting older adults who are taking medications that have been associated with an increased risk of falling. Although the ITT analyses revealed no significant reduction in the rate of recurrent falls, injurious falls, or overall use of high-risk medications, individuals in the intervention group were more likely than those in the control group to discontinue use of a high-risk medication or have the dosage reduced

during the 1-year follow-up period. More work is needed to increase participation rates, increase patient and provider acceptance of the intervention, and evaluate the intervention using a larger sample size that provides adequate power to detect clinically meaningful effects of reduction in the use of high-risk medications on preventing or reducing falls in the high-risk population.

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